5.2 Continuity

Proposed Definition. *Let $E \subseteq \mathbb{R}$ and let $f: E \to \mathbb{R}$. We say that f is continuous when for every $a \in E$*

$$\lim_{x \to a} f(x) = f(a)$$

In terms of our definition of limits, this says:

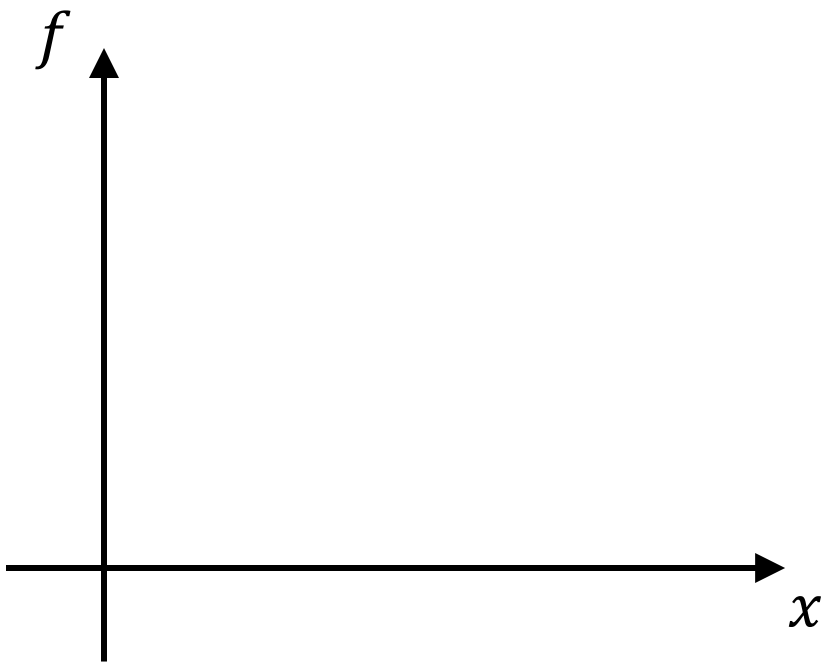
Official Definition 5.13. *Let $E \subseteq \mathbb{R}$, $f: E \to \mathbb{R}$ a function, and $a \in E$. We say that f is <u>continuous</u> at a if, for every $\epsilon > 0$, there exists $\delta > 0$ such that if $x \in E$ and $|x - a| < \delta$ then $|f(x) - f(a)| < \epsilon$.*

We say f is <u>continuous</u> on E if f is continuous at every point $a \in E$.

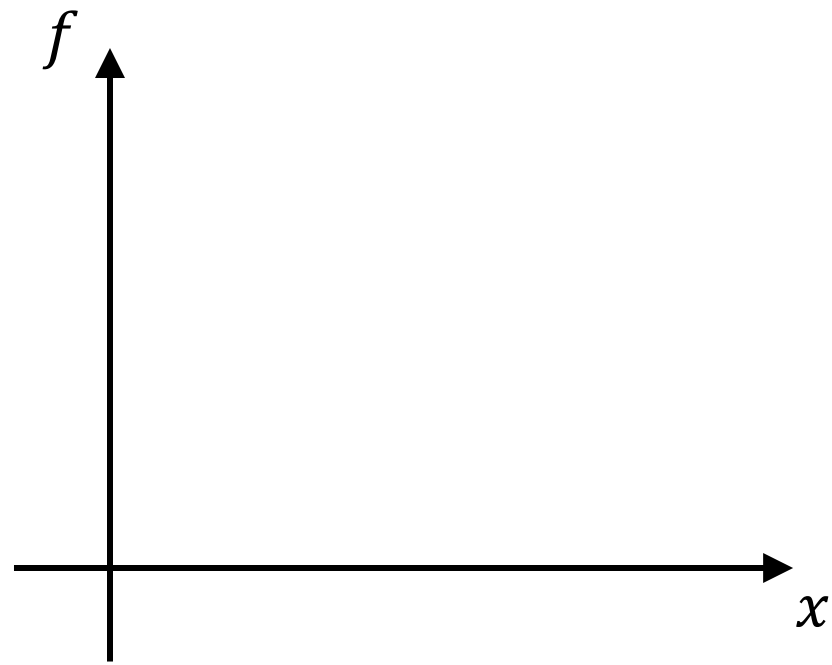
Example. Consider $f(x) = x^2$ at $x = 2$.

$f(x)$ is continuous at $x = 2$ means for any choice of $\epsilon > 0$ we can find $\delta > 0$ such that if $2 - \delta < x < 2 + \delta$ then $4 - \epsilon < x^2 < 4 + \epsilon$. Consider $\epsilon = 1/2$.

Bad choice of δ



Good choice of δ



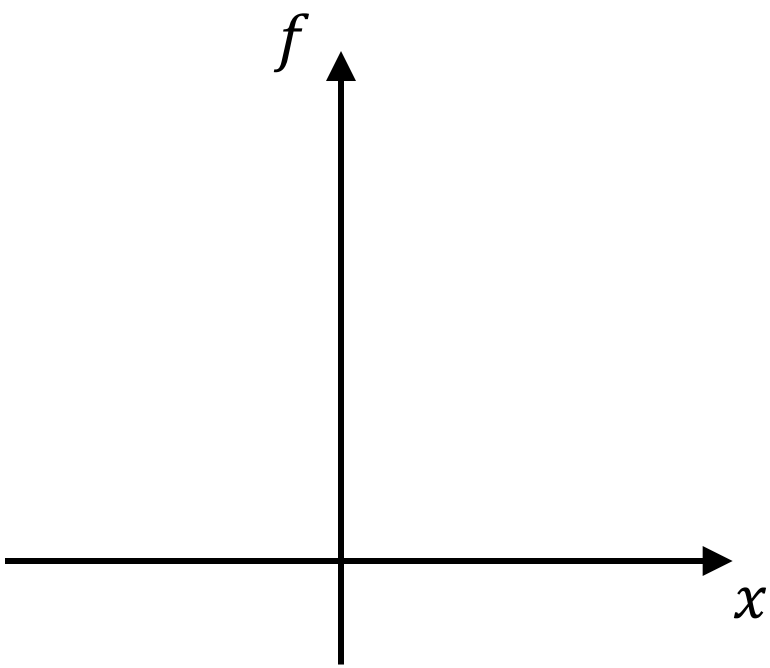
Example 5.14. Show that $f(x) = x$ is continuous on $\mathbb{R}$.

Example 5.15. Show that $f(x) = 1$ is continuous on $\mathbb{R}$.

Let $E \subseteq \mathbb{R}$, $f: E \to \mathbb{R}$ a function, and $a \in E$. f is not continuous at a when

Example 5.16. Show that $f(x)$ is not continuous at $x = 0$ where

$$f(x) = \begin{cases} 2x + 1 & \text{if } x \neq 0 \\ 5 & \text{if } x = 0 \end{cases}$$



Theorem 5.17. *Let $E \subseteq \mathbb{R}$, and $a \in E$ be a limit point of E. Let $f: E \to \mathbb{R}$. The function f is continuous at a if and only if*

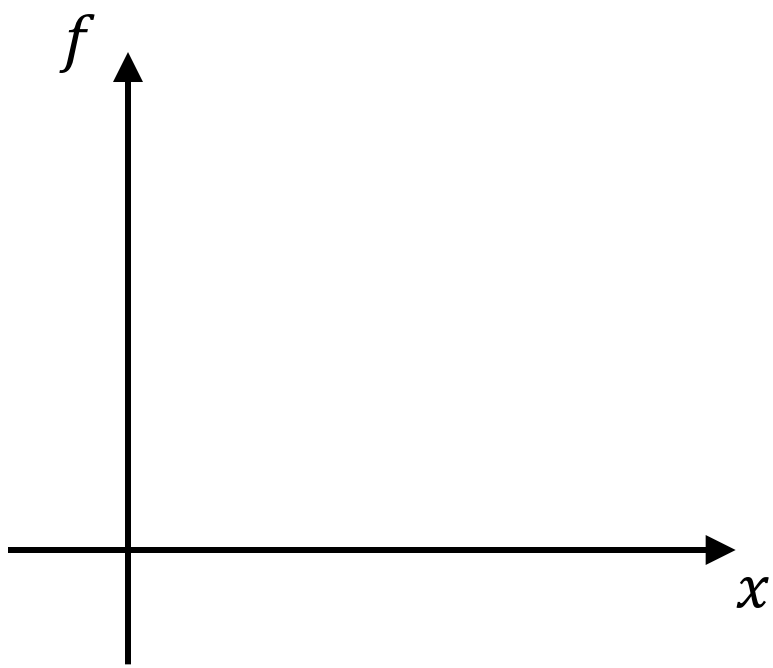
$$\lim_{x \to a} f(x) = f(a).$$

Aside. Our definition of continuity at a does <u>*not*</u> require a to be a limit point of E.

If a is not a limit point of E, then for $\delta > 0$ small, the only $x \in E$ with $|x - a| < \delta$ is $x = a$. Hence $|f(x) - f(a)| = 0 < \epsilon$.

Therefore:

If a is not a limit point of E, then f must be continuous at a.



Theorem 5.18. (Algebra of Continuous Functions)

Let $E \subseteq \mathbb{R}$, $f, g: E \to \mathbb{R}$, *and* $a \in E$. *If* $f(x)$ *and* $g(x)$ *are both continuous at* a, *then*

1. $k\, f(x)$ *is continuous at* a *for all* $k \in \mathbb{R}$;
2. $f(x) + g(x)$ *is continuous at* a;
3. $f(x) \cdot g(x)$ *is continuous at* a;
4. $f(x)/g(x)$ *is continuous at* a *provided* $g(a) \neq 0$.

Corollary 5.20. *For every $n \in \mathbb{N}$, every polynomial of degree n*

$$h(x) = c_0 + c_1 x + c_2 x^2 + \cdots + c_n x^n$$

is continuous on $\mathbb{R}$.

Theorem 5.21. *The following functions are continuous:*

1. $e^x, \cos(x), \sin(x)$ *are continuous on* $\mathbb{R}$;
2. $\log(x)$ *is continuous on* $(0, \infty)$;
3. $x^{1/p}$ *is continuous on* $(0, \infty)$ *for each* $p \in \mathbb{N}^+$.
4. $c_0 + c_1 x + c_2 x^2 + \cdots + c_n x^n$ *is continuous on* $\mathbb{R}$.

Note. For this theorem to make sense, we need to define $e^x, \cos(x), \sin(x)$ etc. One approach involves power series, which we will discuss later in the course.

Theorem. *Let* $A, B \subseteq \mathbb{R}$. *Let* $f: A \to \mathbb{R}$ *and* $g: B \to \mathbb{R}$ *be functions s.t* $f(A) \subseteq B$. *If* $a \in A$ *such that* $f(x)$ *is continuous at* a *and* $g(x)$ *be continuous at* $f(a)$, *then* $g(f(x))$ *is continuous at* a.

Now we want to prove the following facts about a continuous function $f: [a, b] \to \mathbb{R}$.

Theorem 5.25. *If f is continuous on $[a, b]$, then f is bounded on $[a, b]$.*

Theorem 5.28 (Extreme Value Theorem).
If f is continuous on $[a, b]$, then there exists $c, d \in [a, b]$ such that

$$f(c) \leq f(x) \leq f(d) \text{ for all } x \in [a, b].$$

Theorem 5.31. (Intermediate Value Theorem)
If f is continuous on $[a, b]$ and $f(a) < y < f(b)$, then there exists $c \in [a, b]$ such that $f(c) = y$.

Definition 5.22 and 5.23. *Let* $a \in \mathbb{R}$ *and* $\delta > 0$.

The <u>*open*</u> δ-<u>*neighborhood*</u> *of* a *is the interval* $(a - \delta, a + \delta)$.

The <u>*closed*</u> δ-<u>*neighborhood*</u> *of* a *is the interval* $[a - \delta, a + \delta]$.

Definition. *An* <u>*interval*</u> *is a subset of* $\mathbb{R}$ *of the form*

$$(a, b),\ (a, b],\ [a, b),\ [a, b],\ (a, \infty),\ [a, \infty),\ (-\infty, b),\ (-\infty, b],\ \textit{or}\ \mathbb{R}$$

Definition 5.24. *Let $E \subseteq \mathbb{R}$ and $f : E \to \mathbb{R}$ be a function.*
Let $I \subseteq E$ be an interval. Let $A_f = \{ f(x) \mid x \in I \}$.

We say that f is <u>bounded above</u> on I if A_f is bounded above in $\mathbb{R}$.
We say that f is <u>bounded below</u> on I if A_f is bounded below in $\mathbb{R}$.
We say that f is <u>bounded</u> on I if f is both bounded below and bounded above on I.

Lemma 5.26. *Let* $f\colon [a,b] \to \mathbb{R}$ *such that* f *is continuous on* $[a,b]$.
For every $c \in (a,b)$ *there exists* $\delta > 0$ *such that* f *is bounded on the closed* δ*-neighborhood* $[c-\delta, c+\delta]$.

Proof. Let $f\colon [a,b] \to \mathbb{R}$ be continuous on $[a,b]$. Let $c \in (a,b)$.

By a similar argument:

Lemma 5.27. *Let* $f: [a, b] \to \mathbb{R}$ *such that* f *is continuous on* $[a, b]$.
There exists $\delta_a, \delta_b > 0$ *such that* f *is bounded on the closed intervals* $[a, a + \delta_a]$ and $[b - \delta_b, b]$.

We will also need:

Lemma. *Let* $f: [a, b] \to \mathbb{R}$ *such that* f *is continuous on* $[a, b]$.
Let $I, J \subseteq [a, b]$ *are intervals. If* f *is bounded on* I *and* f *is bounded on* J*, then* f *is bounded on* $I \cup J$.

Theorem 5.25. *Let $f\colon [a,b] \to \mathbb{R}$. If f is continuous on $[a,b]$, then f is bounded on $[a,b]$.*

Proof. Let $f\colon [a,b] \to \mathbb{R}$. If f is continuous on $[a,b]$. Let

We want to show that:

1. $\gamma \in B$ i.e. f is bounded on $[a,\gamma]$.
2. $\gamma = b$ i.e. f is bounded on $[a,b]$.

Claim 1. $\gamma \in B$

Claim 2. $\gamma = b$

Theorem 5.28 (Extreme Value Theorem).
If f is continuous on $[a, b]$, then there exists $c, d \in [a, b]$ such that

$$f(c) \leq f(x) \leq f(d) \text{ for all } x \in [a, b].$$

Theorem 5.28 (Extreme Value Theorem).
If f is continuous on $[a, b]$, then there exists $c, d \in [a, b]$ such that

$$f(c) \leq f(x) \leq f(d) \text{ for all } x \in [a, b].$$

Proof. Let f be continuous on $[a, b]$. We want to show that f attains its maximum value at some $d \in [a, b]$. (The proof that c exists is similar.) Let

Minimizing sequence.

Compactness.

Continuity.

Conclusion.

Theorem 5.31. (Intermediate Value Theorem)
If f is continuous on $[a, b]$ and $f(a) < y < f(b)$, then there exists $c \in [a, b]$ such that $f(c) = y$.

Theorem 5.29. *There exists $x \in \mathbb{R}$ such that $x^2 - 2 = 0$.*

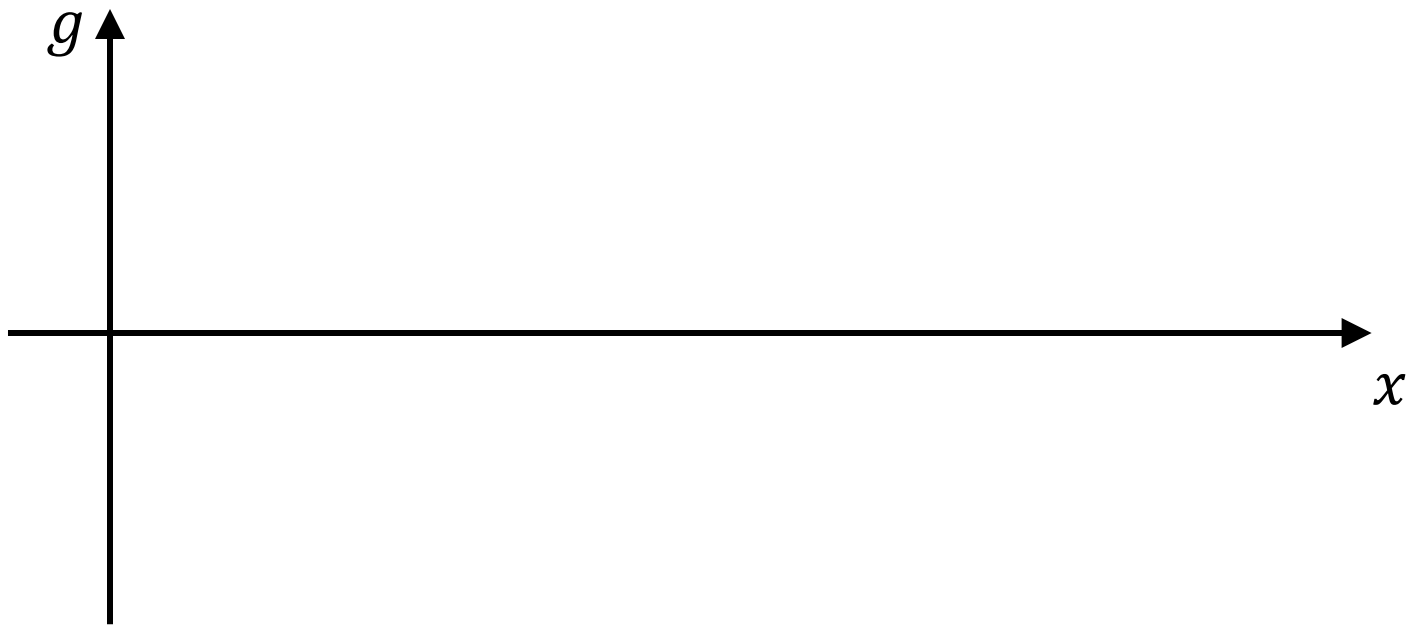
Let $g(x) = x^2 - 2$. Let X be the set of points $x \in [1,2]$ such that $g \leq 0$ on $[1, x]$.

Let $g(x) = x^2 - 2$. Let X be the set of points $x \in [1,2]$ such that $g \leq 0$ on $[1, x]$.

Let $\gamma = \sup X$. We want to show that $g(\gamma) = 0$. Suppose instead that $g(\gamma) > 0$.

By the continuity of $g(x)$ at γ:

For every $\epsilon > 0$ *there exists* $\delta > 0$ *s.t. if* $|x - \gamma| < \delta$ *then* $|g(x) - g(\gamma)| < \epsilon$.



Theorem 5.29. *There exists $x \in \mathbb{R}$ such that $x^2 - 2 = 0$.*

Proof. Let $g(x) = x^2 - 2$ for all $x \in \mathbb{R}$.

We claim that $g(\gamma) = 0$. Notice that since $\gamma \in [1,2]$, g is continuous at γ.

Case 1. Suppose that $g(\gamma) > 0$.

Case 2. Suppose that $g(\gamma) < 0$.

Looking at our proof, let us make the following changes:

- Replace $g(x) = x^2 - 2$ with any continuous function g (relabeled f),
- Replace 1 with any $a \in \mathbb{R}$ such that $g(a) < 0$.
- Replace 2 with any $b \in \mathbb{R}$ such that $g(b) > 0$.

Then by the same proof:

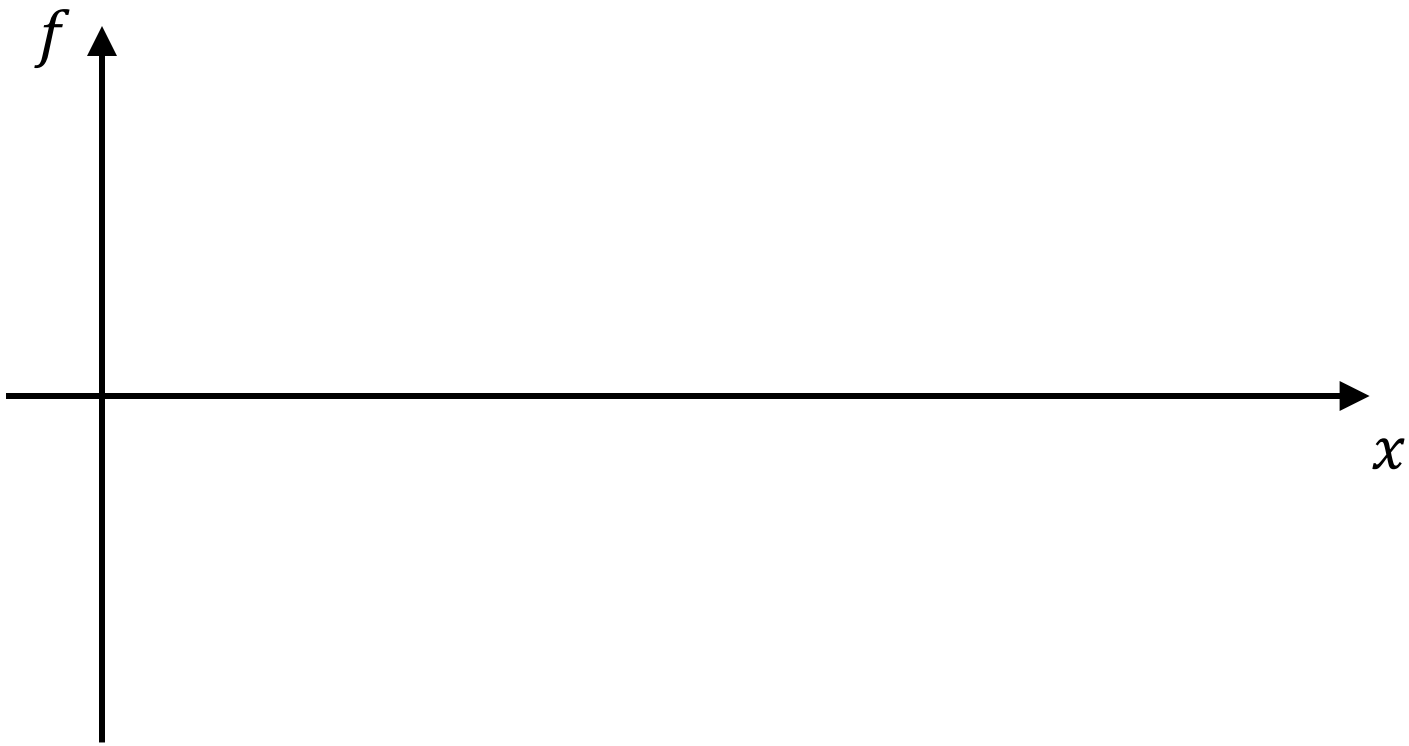
Lemma 5.30. *Let f be continuous on $[a, b]$. If $f(a) < 0$ and $f(b) > 0$, then there exists $c \in [a, b]$ such that $f(c) = 0$.*

Moreover, we showed that:

Since $g(1) = -1$ *and* $g(2) = 2$, *there exists* $c \in [1,2]$ *such that* $g(c) = 0$.

Changing 0 to another value $y \in [-1,2]$ should yield the same result:

For all $y \in [-1,2]$ *there exists* $c \in [1,2]$ *such that* $g(c) = y$.



Theorem 5.31. (Intermediate Value Theorem)
If f is continuous on $[a, b]$ and $f(a) < y < f(b)$, then there exists $c \in [a, b]$ such that $f(c) = y$.

Proof. Let f is continuous on $[a, b]$ and $f(a) < y < f(b)$. Let

Then

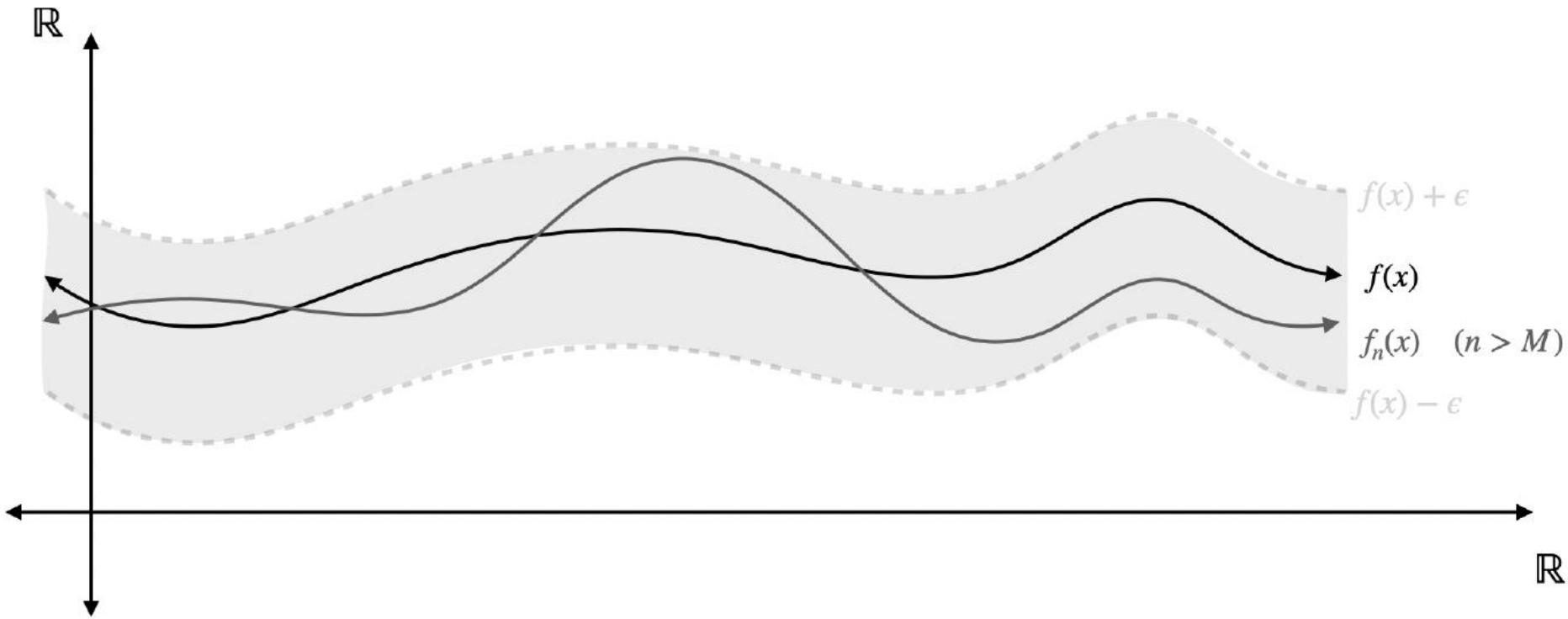
Hence by Lemma 5.30 there exists $c \in [a, b]$ such that $h(c) = 0$. That is

Therefore there exists $c \in [a, b]$ such that $f(c) = y$.

Definition 4.33. *Let $D \subseteq \mathbb{R}$. Let $(f_n(x))$ be a sequence of functions $f_n : D \to \mathbb{R}$ and let $f : D \to \mathbb{R}$ be a function.*

We say that $(f_n(x))$ <u>converges uniformly</u> to $f(x)$ when

$$\forall \epsilon > 0 \; \exists M = M(\epsilon) \in \mathbb{N}^+ \; s.t. \; \forall x \in D \; if \; n > M \; then \; |f_n(x) - f(x)| < \epsilon.$$

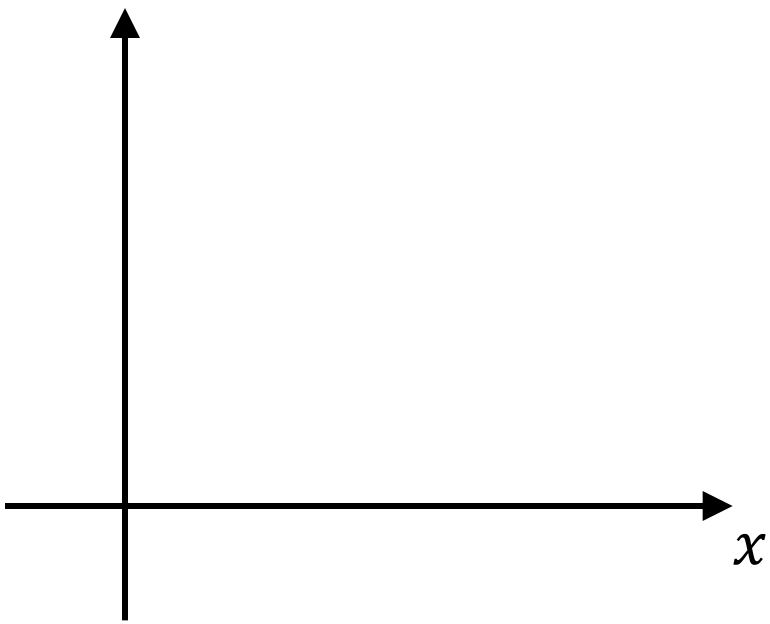


Theorem 5.32. (Uniform Limit Theorem)

Let $(f_n(x))$ *be a sequence of continuous functions on a domain* D.

If $(f_n(x))$ *uniformly converges to* $f(x)$ *on* D*, then* $f(x)$ *is continuous on* D.

Example. $f_n(x) = x^n$ on the domain $[0,1]$.



Theorem 5.32. (Uniform Limit Theorem)

Let $(f_n(x))$ *be a sequence of continuous functions on a domain* D.

If $(f_n(x))$ *uniformly converges to* $f(x)$ *on* D*, then* $f(x)$ *is continuous on* D.